

Bags, Vectors & Transformers

The Limits of Bag-of-Words · Day 1, Part 5

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So Far, So Good — But...

Bag-of-words methods are **workhorses**: transparent, fast, and genuinely useful. They have carried decades of social science research. But the **simplifying assumption** that makes them work — treating a document as an unordered pile of word counts — is also their **core weakness**.

In the next few slides we look at **what bag-of-words throws away**, and why that matters. Understanding these limits is exactly what **motivates embeddings** — the approach we turn to next.

Drawback 1: Word Order Is Lost

A bag-of-words model sees only **counts**, not **sequence**. Two sentences with the same words but opposite meaning look **identical**.

“the dog bit the man”

“the man bit the dog”

Same words. Same counts.

Identical to the model.

But the **meaning is completely different**. Word order carries grammar, agency, negation — and bag-of-words discards all of it.

Partial fixes exist (n-grams capture short sequences), but they explode the vocabulary and still miss longer-range structure.

Drawback 2: No Sense of Meaning

To a bag-of-words model, every word is a **separate, unrelated symbol**. It has **no idea** that some words mean similar things.

“happy” and **“joyful”**

are as unrelated to the model as

“happy” and **“concrete”**

Each is just a separate column in the matrix.

Why it hurts: two documents can be about the **same thing** in different words and look **completely dissimilar** — “car” vs “automobile”, “doctor” vs “physician”.

This is the big one — and the main gap embeddings are designed to close.

Drawback 3: High Dimensionality & Sparsity

The document-term matrix has **one column per word** in the vocabulary. Real corpora have **tens of thousands** of words — so the matrix is **enormous and mostly zeros**.

The problem

- Any single document uses only a **tiny fraction** of the vocabulary
- Most cells are **zero** (sparse)
- Thousands of **dimensions** to work in

Why it hurts

- Computationally **heavy**
- Statistically **harder** to work with
- Similarity measures get **noisy** in high dimensions

*Embeddings represent words in a few hundred **dense** dimensions instead of tens of thousands of sparse ones.*

Drawbacks 4 & 5: New Words & Context

Out-of-vocabulary words

- The model only knows words it **saw during training**
- A **new word** (or typo) is simply **invisible**
- No way to guess meaning from **similar** words

One meaning per word

- “**bank**” (river) and “**bank**” (money) are the **same column**
- Bag-of-words **cannot tell them apart**
- Context is ignored entirely

Static embeddings help with the first; contextual embeddings (BERT) solve the second.

Why This Points to Embeddings

Every drawback traces back to **one root problem**: bag-of-words treats words as **isolated symbols** with no relationship to each other. What if, instead, we could represent each word as a **position in space** — where **similar words sit close together**?

Bag-of-words

- Words are **separate symbols**
- Sparse, high-dimensional
- No notion of similarity
- Order and context lost

Embeddings

- Words are **dense vectors**
- Compact, few hundred dimensions
- **Similar words sit close together**
- Context can be captured (later)

That is exactly the idea we turn to next: word embeddings.